

67 Theorem

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Theorem Statement

For any positive integer n except 1, 2, 10, or 11, the digits of n can be rearranged to form the number 67.

For example: Given integer 3,231,490, we can perform the following operations,

$(3*2) = 6$ and $(3-1) + (-4+9) + 0 = 7$, then we have digits 6 and 7. 67!

Subproblem: Rearranging to 6

Theorem: For any integer string x (no leading zeros) except "1", "2", "10", or "11", the digits of x can be rearranged to form the digit string "6".

Proof Strategy: Strong induction on the length $\ell = |x|$ of the string x . For the induction step, assume $x = wa$ where w has length $\ell - 1$ and $a \in \{0, 1, \dots, 9\}$. By the induction hypothesis, since $|w| < \ell$, the digits of w rearrange to "6". Thus, the digits of x rearrange to some permutation of "6" followed by a , denoted " $6a$ ". It remains to show that for every $a \in \{0, \dots, 9\}$, the digits of " $6a$ " rearrange to "6" via mathematical identities, proved by exhaustion below.

Connection to 67: Given a rearrangement to "6", append "7" via the digit-preserving identity $6! = 720$, where 720 rearranges to 7 2 0! (since $0! = 1$), then to 7 2 1, then $7(2+1) = 7\ 3$, then $7\ 3! = 7\ 6$, then 67. Thus, the subproblem implies the main theorem.

Base Cases

Verify single-digit, two-digit, and three-digit strings directly (proof by exhaustion):

- Single-digit: 3–9 rearrange to "6":

x	Rearrangement Sequence to "6"
3	$3! = 6$
4	$4! = 24 \rightarrow 2+4 = 6$
5	$5! = 120 \rightarrow 1+2+0 = 3 \rightarrow 3! = 6$
6	Trivial
7	$7! = 5040 \rightarrow 5+(0+4+0)! = 6$
8	$8! = 40320 \rightarrow 4+(0*3)+2+0 = 6$
9	$\sqrt{9} = 3 \rightarrow 3! = 6$

Exceptions: 1, 2 lack digits for "6".

- Two-digit: 12–19, 20–29, cases where cannot use prior bases (1, 2) to perform the induction, rearrange to "6"

x	Rearrangement Sequence to "6"
12	$1+2 = 3 \rightarrow 3! = 6$
13	$1*3 = 3 \rightarrow 3! = 6$
14	$4-1 = 3 \rightarrow 3! = 6$
15	$1+5 = 6$
16	$1*6 = 6$
17	$7-1 = 6$
18	$1+8 = 9 \rightarrow \sqrt{9} = 3 \rightarrow 3! = 6$
19	$1*9 = 9 \rightarrow \sqrt{9} = 3 \rightarrow 3! = 6$
20	$2+0! = 3 \rightarrow 3! = 6$
21	$2+1 = 3 \rightarrow 3! = 6$
22	$2+2 = 4 \rightarrow 4! = 24 \rightarrow 2+4 = 6$
23	$2*3 = 6$
24	$2+4 = 6$
25	$2+5 = 7 \rightarrow 7! = 5040 \rightarrow 5+(0+4+0)! = 6$
26	$2*6 = 12 \rightarrow 1+2 = 3 \rightarrow 3! = 6$
27	$2*7 = 14 \rightarrow 4-1 = 3 \rightarrow 3! = 6$
28	$8-2 = 6$
29	$2*9 = 18 \rightarrow 1+8 = 9 \rightarrow \sqrt{9} = 3 \rightarrow 3! = 6$

Exceptions: 10, 11 lack digits for "6".

- Three-digit: 100-109, 110-119, cases where cannot use prior bases (10, 11) to perform the induction, rearrange to "6"

x	Rearrangement Sequence to "6"
100	$1+0!+0! = 3 \rightarrow 3! = 6$
101	$1+0!+1 = 3 \rightarrow 3! = 6$
102	$1+0+2 = 3 \rightarrow 3! = 6$
103	$(1+0)*3 = 3 \rightarrow 3! = 6$
104	$4-1+0 = 3 \rightarrow 3! = 6$
105	$1+0+5 = 6$
106	$(1+0)*6 = 6$
107	$7-1+0 = 6$
108	$1+0+8 = 9 \rightarrow \sqrt{9} = 3 \rightarrow 3! = 6$
109	$(1+0)*9 = 9 \rightarrow \sqrt{9} = 3 \rightarrow 3! = 6$
110	$1+1+0! = 3 \rightarrow 3! = 6$
111	$1+1+1 = 3 \rightarrow 3! = 6$
112	$1+1+2 = 4 \rightarrow 4! = 24 \rightarrow 2+4 = 6$
113	$(1+1)*3 = 6$
114	$1+1+4 = 6$
115	$1+1+5 = 7 \rightarrow 7! = 5040 \rightarrow 5+(0+4+0)! = 6$
116	$(1+1)*6 = 12 \rightarrow 1+2 = 3 \rightarrow 3! = 6$
117	$(1+1)*7 = 14 \rightarrow 4-1 = 3 \rightarrow 3! = 6$
118	$8-1-1 = 6$
119	$(1+1)*9 = 18 \rightarrow 1+8 = 9 \rightarrow \sqrt{9} = 3 \rightarrow 3! = 6$

Induction Hypothesis

Assume that any integer string w can be rearranged to form the digit string "6"; $|x| \geq |w|$.

Inductive Step (Proof by Exhaustion on a)

For $x=wa$, digits form "6 a " multiset. Exhaust cases $a=0$ to 9, showing each rearranges to "6":

a	Rearrangement Sequence to "6"
0	$6\ 0 \rightarrow 6+0 = 6$
1	$6\ 1 \rightarrow 6*1 = 6$
2	$6\ 2 \rightarrow 6*2 = 12 \rightarrow 1+2 = 3 \rightarrow 3! = 6$
3	$6\ 3 \rightarrow 6-3 = 3 \rightarrow 3! = 6$
4	$6\ 4 \rightarrow 6*4 = 24 \rightarrow 2+4 = 6$
5	$6\ 5 \rightarrow 6*5 = 30 \rightarrow 3+0 = 3 \rightarrow 3! = 6$
6	$6\ 6 \rightarrow 6+6 = 12 \rightarrow 1+2 = 3 \rightarrow 3! = 6$
7	Trivial*
8	$6\ 8 \rightarrow 6+8 = 14 \rightarrow 4-1 = 3 \rightarrow 3! = 6$
9	$6\ 9 \rightarrow 6+9 = 15 \rightarrow 1+5 = 6$

*Trivial to the superproblem

Conclusion

Each case uses digit-preserving operations (factorials, sums, subtractions, multiplication, square root within multiset), closing to "6".

By strong induction, the subproblem holds. Thus, the theorem follows.

Therefore, for any positive integer n except 1, 2, 10, or 11, the digits of n can be rearranged to form the number 67.